Test XB_0031 P2 Final Exam

Test ID: 61678
Folder: /Top/SIS/XB_0031
Folder description: Intelligent Systems
Version: 6.0
Randomised: No
Last modified: Saturday, 19 december 2020 13:18:00
Number of questons: 23
Blocks: Fixed
Display questions once: No
Tools: Calculator extended
Test time: 105 minutes
Maximum score: 62 pt.
Chance score: 10.80 pt. / 17%
In test set with: -

Proctoring

Question order: Fixed

Question 1 – Proctoring rules – 219297.1.1

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021
Answer option order: Random
Partial scoring: No
Maximum score: 1
Chance score: 0.50 pt. / 50%
Status:
Last modified: 17/12/2020 01:24
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

OLD CONTENT

Please make sure that you follow the Proctoring protocol. This means that at this stage

- you should have done **the desk-scan** (not a room scan, but shown your desk). You can still do it at this stage.
- **only permitted tools** are on your desk (Calculator, scrap paper).
- you are **clearly visible** (and remain so during the exam)
- you will **not communicate** with anybody during the exam (i.e. phones etc are OFF and not in proximity).
- you will have shown your **student-id**, which clearly shows your image
- you do not use **head-phones**

A I did it and fullfill all the requirements of the protocol.

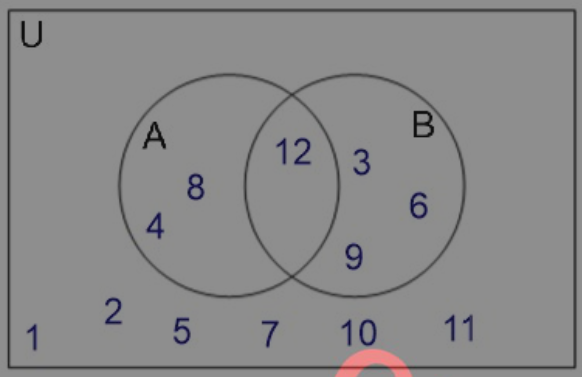
B I did not do it. I am aware of the risk that my exam might be declared invalid.

Set Theory

Question order: Fixed

Question 2 – Venn diagram 2 – 217973.2.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions set theory, relations and vagueness
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
Last modified: 17/12/2020 01:20
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*



Given the Venn diagram above, what is the complement of A union B, (A U B)'?

OLD CONTENT

A {12}

B {1,2, 3, 4, 5, 6, 7, 8, 9, 10 ,11}

C {1, 2, 5, 7, 10, 11}

D {1, 2, 5, 7, 10 ,11, 12}

Question 3 – Cardinality 3 – 217107.2.0

Question type: Fill in (numerical)
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions set theory, relations and vagueness
Answer option order: Fixed
Partial scoring: No
Maximum score: 2
Chance score: 0.00 pt. / 0%
Status:
Last modified: 17/12/2020 00:18
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Suppose we have a universal set that contains 41 elements. If |A| = 14, |B| = 19 and |A ∪ B| = 28, what is the cardinality of A intersection B, |A ∩ B|?

5

Uncertainty and Vagueness

Question order: Fixed

Question 4 – Uncertainty versus Vagueness (dutch man) – 205072.1.1

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions set theory, relations and vagueness
Answer option order: Random
Partial scoring: No
Maximum score: 1
Chance score: 0.50 pt. / 50%
Status:
Last modified: 17/12/2020 20:14
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following statement: "Half of the Dutch men are married". Which of the following statements is true?

A The author of this statement describes the marital status of Dutch men with a degree of vagueness.

B The author of this statement describes the marital status of Dutch men with a degree of uncertainty.

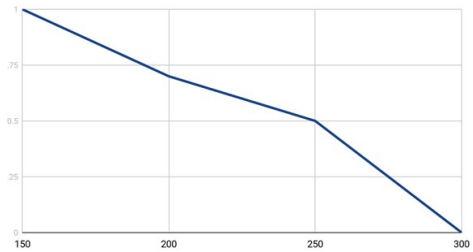
Question 5 – Fuzzy definitions – 219971.1.1

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions set theory, relations and vagueness
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.67 pt. / 33%
Status:
Last modified: 17/12/2020 00:04
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

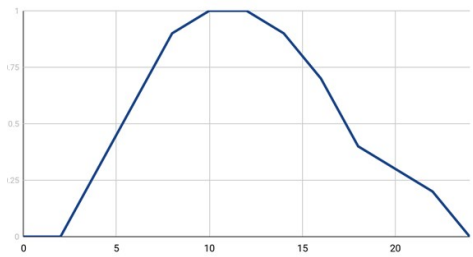
The director of our institute wants to reduce the environmental impact of the company by reducing the CO2 emissions caused by transportation. Therefore, she wants to give a devise a policy to support those who do not live very close (they can come by bicycle) or those who live very far (they should do remote work).

Since you know the director is a big fan of fuzzy set theory, you decide to model this problem with fuzzy sets. How do you model the fuzzy set of "**employees who live moderately far away**", i.e. those who do not live close, but also not far away?

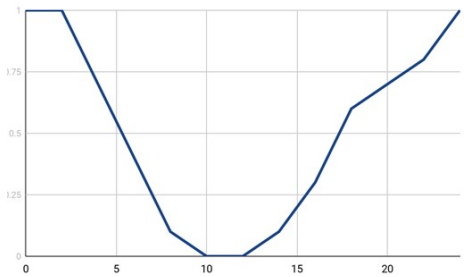
A



B



C



Question 6 – Fuzzy Logic 2 – 217115.3.0

Question type: Fill in (multiple)
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions set theory, relations and vagueness
Answer option order: Fixed
Partial scoring: Yes
Maximum score: 4
Chance score: 0.00 pt. / 0%
Status:
Last modified: 17/12/2020 01:14
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Given the following rules for fuzzy logic:

– f(¬A) = 1 - f(A)
– f(A v B) = max(f(A);f(B))
– f(A & B) = min(f(A);f(B))
– f(A -> B) = min(1;1 – f(A) + f(B))

In Fuzzy logic the meaning of a formula is defined as a value between 0 and 1. What is the meaning of f(((A -> B) v B) & (¬A v ¬B)) if f(A)=0.8 and f(B)=0.5?

f((A -> B) v B) = 0.7
f(¬A v ¬B) = 0.5
f(((A -> B) v B) & (¬A v ¬B)) = 0.5

[Alphanumeric]	[Alphanumeric]	[Alphanumeric]
0.7	**0.5**	**0.5**
o,7	0,5	0,5

Probability Theory

Question order: Fixed

Question 7 – Full joint probability distribution – 217964.3.0

Question type: Multiple response
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions set theory, relations and vagueness
Answer option order: Random
Partial scoring: Yes
Maximum score: 3
Chance score: 0.89 pt. / 30%
Status:
Last modified: 17/12/2020 00:24
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following full joint probability distribution:

P	Q	R	P(P, Q, R)
true	true	true	0.26
true	true	false	0.33
true	false	true	0.04
true	false	false	0.14
false	true	true	0.03
false	true	false	0.08
false	false	true	0.11
false	false	false	0.01

Which statements are true?

A P(R = false) = 0.56

B P(P = true v Q =true v R = true) = 1

C P(P = true & Q = true) = 0.59

D P(R = false) = 0.63

E P(Q = true) = 0.26

F P(Q=true|R-true) = P(R=true|Q=true)

Question 8 – Bayes Rule – picnic – 217977.3.0

Question type: Fill in (numerical)
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions set theory, relations and vagueness
Answer option order: Fixed
Partial scoring: No
Maximum score: 4
Chance score: 0.00 pt. / 0%
Status:
Last modified: 17/12/2020 01:10
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

You are planning a picnic today, but the morning is cloudy
- 50% of all rainy days start off cloudy.
- But cloudy mornings are common, as about 40% of days start cloudy.
- This month is usually a dry month (only 10% of the days tend to be rainy)

What is the chance of rain during the day given the morning is cloudy?

0.125

12.5

Question 9 – Bayesian Networks theory – 219972.1.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions set theory, relations and vagueness
Answer option order: Random
Partial scoring: No
Maximum score: 1
Chance score: 0.25 pt. / 25%
Status:
Last modified: 17/12/2020 00:13
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

What is the most important assumption that underlies the idea of Bayesian Networks, which allows these models to calculate probabilites for large sets of random variables efficiently?

A The naiveness assumption that all variables are independent, and that the denominator in Bayes rule can be ignored. Therefore we only deal with estimates and not proper probabilites.

B The assumption that variables are independent of their siblings and other non-descendend nodes, so that their probability can be calculated only on the basis of their parents.

C The assumption that from unsatisfiable formulas everything can be derived. This means that the probability of a child is conditional of the probability of their parents and their siblings in the network.

D The assumption that there are only finitely many models, and that we can therefore simply add the conditional probabilities in the Bayesian Networks to calculate the local probabilities.

Question 10 – Bayesian Networks – Mary Calls – 217962.9.0

Question type: Fill in (numerical)
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions set theory, relations and vagueness
Answer option order: Fixed
Partial scoring: No
Maximum score: 4
Chance score: 0.00 pt. / 0%
Status:
Last modified: 17/12/2024 21:38
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

B	P(B)
+b	0.002
-b	0.998

E	P(E)
+e	0.001
-e	0.999

B	E	A	P(A|B,E)
+b	+e	+a	0.92
+b	+e	-a	0.08
+b	-e	+a	0.90
+b	-e	-a	0.10
-b	+e	+a	0.31
-b	+e	-a	0.69
-b	-e	+a	0.002
-b	-e	-a	0.998

A	J	P(J|A)
+a	+j	0.7
+a	-j	0.3
-a	+j	0.01
-a	-j	0.99

A	M	P(M|A)
+a	+m	0.8
+a	-m	0.2
-a	+m	0.001
-a	-m	0.999

Given the following Bayesian network, what is the probability that Mary calls? Round your answer to 6 decimal places.

0.004279

0.0042 - 0.0044

Machine Learning

Question order: Fixed

Question 11 – Learning Tasks – 205101.2.1

Question type: Fill in (multiple)
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Fixed
Partial scoring: Yes
Maximum score: 3
Chance score: 0.00 pt. / 0%
Status:
Last modified: 17/12/2020 14:02
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

What kind of model do we need for the following cases (Classification, Regression, Ranking, Clustering)? Also indicate whether this would be supervised or unsupervised machine learning.

Please write words (from the above lists, without spaces) into the appropriate fields.

- A model that predicts the temperature based on air pressure and light intensity. We trained it with a lot of past observations.
  - Kind of model? Regression
  - Supervised/Unsupervised? supervised
- We want to make a system that sorts pictures according to what kind of landscape they represent (One of forest, city, dessert, coastline, and others). We used a lot of examples to train the system.
  - Kind of model? Classification
  - Supervised/Unsupervised? supervised
- We want to create groups of researchers working on similar topics, what ML technique do we use?
  - Kind of model? Clustering
  - Supervised/Unsupervised unsupervised
- We want to order search results. We do this based on how often the search term occurs on the page.
  - Kind of model? Ranking
  - Supervised/Unsupervised unsupervised

[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Alphanumeric]
Regression	**supervised**	**Classification**	**supervised**	**Clustering**
[Alphanumeric]	[Alphanumeric]	[Alphanumeric]		
unsupervised	**Ranking**	**unsupervised**		

Question 12 – Basic Machine Learning Evaluation – 205084.1.2

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Random
Partial scoring: No
Maximum score: 1
Chance score: 0.25 pt. / 25%
Status:
Last modified: 17/12/2020 20:13
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Which of the following statements is true?

A Evaluating with cross validation is often not applicable, as it requires too much data

B Evaluating on the training data will give untrustworthy results

C Evaluating on a specific testing set will result in overfitting

D Evaluating a Machine Learning model should always done by people who were not involved in the learning

Question 13 – Classification matrix 1 – 217100.2.1

Question type: Fill in (multiple)
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Fixed
Partial scoring: Yes
Maximum score: 3
Chance score: 0.00 pt. / 0%
Status:
Last modified: 17/12/2020 18:57
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

What is the Precision and Recall for this classification (no rounding)?

	actual +	actual -
predicted +	525	75
predicted -	225	340

Precision: 0.875
Recall: 0.7

[Numeric]	[Numeric]
0.875	**0.7**

Question 14 – Design Matrix – 218341.4.2

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Random
Partial scoring: No
Maximum score: 3
Chance score: 0.60 pt. / 20%
Status:
Last modified: 19/12/2020 17:03
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following training set of positive examples examples {(1,1),(1,-1),(-2,2),(-2,-2)} and negative examples {(3,3),(3,-3),(-4,4),(-4,-4)}. As seen in the last working sheet there is no linear separator (a linear function separating all the positive from the negative examples).

Consider a design matrix with two new features (replacing the old ones) $z_1 = x*y$ and $z_2 = x+y$.

Which of the following statements is true?

A The function $z_1 = 2* z_2 + 2$ is a linear separator for this new design matrix.

B There is still no linear separator for this new design matrix

C There are too many linear separators for this new design matrix to decide.

D The function $z_1 = - z_2 + 4$ is a linear separator for this new design matrix

E Neither $z_1 = 2* z_2 + 2$ nor $z_1 = - z_2 + 4$ are linear separators for this design matrix, but there exist one.

Question 15 – Bayesian Learning: Reading – 205096.2.2

Question type: Multiple response
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Random
Partial scoring: Yes
Maximum score: 4
Chance score: 1.19 pt. / 30%
Status:
Last modified: 17/12/2020 20:08
Attributes: *Moeilijkheid/Difficulty* *01 Gemiddeld/Average*
Taxonomie/Taxonomy *03 Toepassen/Apply*

Consider the trading agent trying to infer which books a user reads based on keywords supplied for earlier books he read. Suppose the learning agent has the following data:

Reading preferences S:					
Books	crime	academic	local	music	**Reads**
b1	True	false	false	true	**true**
b2	True	false	false	false	**false**
b3	false	true	false	true	**true**
b4	false	true	true	true	**false**
b5	false	false	false	true	**true**

Apply a Bayes Learner whether the following book will be read or not:

b6	true	true	false	false

We apply smoothing, which assumes a value 0.01 in case of no occurrences.

Which of the following statements is true?

A P(-reads|+crime,+academic,-local,-music) is estimated through 2/5 * 1/2 * 1/2 * 1/2 * 1/2 = 0.025

B P(-reads|+crime,+academic,-local,-music) is estimated through 3/5 * 1/3 * 1/3 * 2/3 * 0.01 = 1/2250

C P(reads|+crime,+academic,-local,-music) is estimated through 3/5 * 1/3 * 1/3 * 1 * 0.01 = 1/1500

D P(reads|+crime,+academic,-local,-music) is estimated through 2/5 * 1/2 * 1/2 * 1/2 * 1/2 = 0.025

E P(reads|+crime,+academic,-local,-music) is estimated through 2/5 * 1/2 * 0 * 1/2 * 1/2 = 0

F The algorithm predicts that the reader will probably read the book *b6*.

Question 16 – Text Features – 219973.1.1

Question type: Multiple response
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Random
Partial scoring: Yes
Maximum score: 2
Chance score: 0.60 pt. / 30%
Status:
Last modified: 17/12/2020 18:59
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the sentence:

In the true Computer Sciences, where science rules, a rule computes truth.

Which operations have been applied to this sentence when it has been reduced to the following set of text features:

< true, computer, science, sciences, rule, rules, computes, truth >

A Stopword removal

B Stemming

C Remove Capitalisation

D Remove Punctuation

E Parsing

F Sentiment Analysis

G Noun Tagging

H Part of Speech Tagging

Question 17 – kNN – 205107.1.1

Question type: Multiple response
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Random
Partial scoring: Yes
Maximum score: 2
Chance score: 0.67 pt. / 33%
Status:
Last modified: 17/12/2020 20:13
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

For K-nearest neighbours, which of the following are true (multiple answer can be correct)?

A it is always best to choose k as big as the available dataset. This improves classification accuracy, because there will be no problems with noise.

B We use the majority label to reduce effects of noise in the dataset.

C We do not need to define how the system measures the distance between two points because it only needs to find the nearest ones.

Question 18 – knn learning – 205103.2.2

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Random
Partial scoring: No
Maximum score: 4
Chance score: 1.00 pt. / 25%
Status:
Last modified: 17/12/2020 20:13
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

You are given the following words extracted from documents (one sentence is a document for this exercise), with the associated label.

1. {dog, sleep, near, field} label: passive
2. {cat, walk, field} label: active
3. {field, is, full, poppy} label: passive
4. {cat, dog, run, field} label: active
5. {cat, is, mad, dog, sleep, poppy} label: passive

Suppose you use a k-nearest neighbour algorithm (k=3) to decide whether the sentence {cat, sleep, poppy, dog} belongs to the class active or passive.

Take the following steps:
1. convert each of these sets into a feature vector (this is what we did in the lecture, but this step is not strictly necessary given the distance function we will use in 2)
2. create a k-NN classifier, with k = 3. As a distance, we use "1/(number of words in common)" --- / stands for devided by.
3. Use a majority voting to decide the label. In case of a tie, increase k, until the tie breaks.

Which of the following statements is true?

A The three nearest neighbours are **1, 4, 5** with labels **passive, active, passive** so the classification is **passive**.

B The three nearest neighbours are **1, 2, 5** with labels **passive, active, passive** so the classification is **passive**.

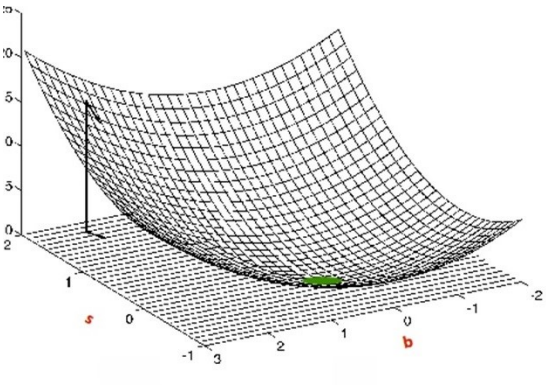
C The three nearest neighbours are **1, 2, 4** with labels **passive, active, active** so the classification is **active**

D The three nearest neighbours are **2, 4, 5** with labels **active, active, passive** so the classification is **active.**

Question 19 – Model space MC – 205100.3.0

Question type: Multiple response
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Random
Partial scoring: Yes
Maximum score: 5
Chance score: 1.53 pt. / 31%
Status:
Last modified: 20/12/2023 16:56
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

In the following image, also shown in the lecture, the green point is identified as the optimal solution to a learning problem.



Which of the following statements are true? (multiple answers are possible).

A The green point refers refers to the linear function with the lowest error rate (the one that fits a particular learning model best according to the training data)

B The green point refers to the point in the binary vectorspace with the best training example (the one that fits a particular learning model best)

C The third dimension (orthogonal to s and b) shows the error rate for a linear separator function, eg. f(x) = sx+b.

D The green point is the point where f(x) = sx+b reaches a minimum, ie where the linear separator reaches the lowest point (in other words, f(x)=0

E The third dimension (orthogonal to s and b) enumerates all possible linear separators, so that the green point is the optimal one (as it is the lowest).

F There are two ways to find the best machine learning model in this scenario: either we analytically find the green point as the point where the derivation of the error function is 0, or we apply gradient descent as a search method to move step by step to the green point. (Gradient descent is a local search similar to hill-climbing)

G With a Neural Network we can predict the value of the 3rd dimension (orthogonal to s and b on the x and y axis) given the representation of the features according to variables s and b.

Question 20 – Regression – 219974.4.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Random
Partial scoring: No
Maximum score: 4
Chance score: 0.67 pt. / 17%
Status:
Last modified: 15/12/2021 15:40
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

This exercise is about Regression.

Let us try to fit a linear function to a number of training sets where the slope s is known to be 1.

Suppose we have three pairs of values (1,0), (2,0) and (3,1) for two features. What is the optimal fitting function in this model class?

A f(x)= x - 5/3

B f(x)= x - 3/5

C f(x)= x + 5/3

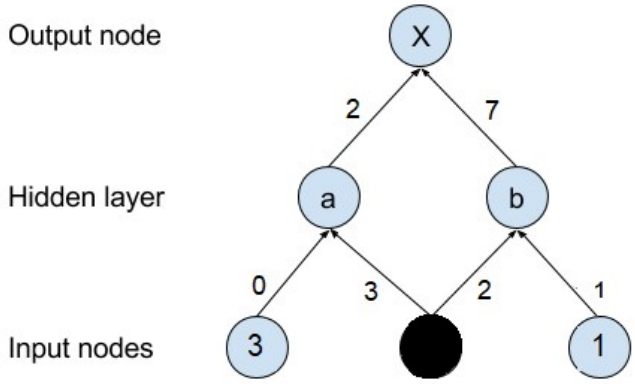
D f(x)= x + 3/5

E There is no optimal fitting function

F Neither of the functions in the other options is the optimal fitting function, but there exist one.

Question 21 – Neural Network – value of an output node – 217098.1.0

Question type: Fill in (numerical)
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Fixed
Partial scoring: No
Maximum score: 3
Chance score: 0.00 pt. / 0%
Status:
Last modified: 10/12/2020 18:57
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*



In the above image, a basic multilayer feedforward neural network, with weights and values on the input nodes is shown. It has no activation function (all nodes are linear). The black node is a bias node. What is the value on the output node (marked X)?

27

Philosophy of Mind

Question order: Fixed
There will be a single question related to Philosophy of Mind.

Question 22 – Symbol Grounding – 218343.1.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Philosophy of Mind
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
Last modified: 14/12/2020 00:05
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

What is meant by "symbol grounding"

A Symbol grounding is the idea that the Chinese characters in Searle's Chinese Room experiment are mapped to English words, so that the person in the box can understand them

B Symbol grounding is the idea that a true understanding of abstract concepts requires a mapping to sensory experience

C Symbol grounding is the idea that all intelligent behaviour is grounded in symbol manipulation

D Symbol grounding is the idea that robots can never be truly intelligent, becaue they are not as grounded as humans are

Question 23 – Turing Test – 218342.1.1

OLD CONTENT

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Philosophy of Mind
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
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Which one of the below is a reasonable objection to the Turing test

A The Turing Test dates from the '50s and computers are very different now

B The Turing Test incorrectly assumes that all intelligence will be similar to human intelligence

C The Turing Test only tests for intelligence, and not for consciousness, which leads to falls negatives

D The Turing Test is limited to narrow domains of expertise